Editor Comments to Author:

Reviewer(s)' Comments to Author:

Reviewer: 1

Comments to the Author
This is a very nice paper/protocol describing the use of ALDEx2 to perform differential expression analysis. The writing is clear and approach and protocol easy to follow. Below are some minor comments mainly aimed at expanding its reach and uptake.

We thank the reviewer for their kind words and for their helpful feedback. Please see below our responses to the comments in blue text.

1. In the introduction, it is worth have a couple of sentences describing approaches that are starting to be used to monitor microbial load (e.g. spike ins, CFUs and flowcell)

This is a great point- we thank the reviewer for mentioning it. We have expanded the first paragraph of the introduction accordingly.

2. Given this seems to be an improved version of ALDEx2, has it reached the stage of development that it should be called ALDEx3?

ALDEx3 is currently in development and, while it has many of the same features as ALDEx2, it is based on linear models and will function slightly differently than its predecessor. As part of our response to point #13, we now mention this in the commentary section.

3. The title is quite specific (ALDEx2 and differential expression) could it be made more generalizable? something along the lines of e.g. "Rigorous testing of differential abundance or expression of genes captured by next generation sequence datasets??"

We thank the reviewer for their suggestion; however, we are aiming to raise awareness of the advantages of using scale models during differential abundance/expression analysis and ALDEx2 is the only tool at present that offers this. Moreover, a title that includes the key feature of this protocol would be better picked up by search engines, given the myriad tools available for differential abundance/expression analysis, so we have opted to keep the title as-is.

4. Throughout please provide version numbers of other software (e.g. Bioconductor, R, RSTudio that used in this protocol)

We thank the reviewer for pointing out this accidental omission. We have now added this information in the 'Materials' section below each protocol.

5. In Basic Protocol 1 step 3, add a note somewhere pointing to the commentary for a description of the gamma parameter (same for protocol 2 step 5 and also where you refer to other parameters described in commentary)

We have now added a note in italic text underneath the corresponding steps pointing to the commentary as requested.

6. Beyond the two protocols provided, could you add a small section explaining what you would do with your own dataset

We initially left any further analysis or visualisation out of the protocol as each analyst will have their own preferred approach, but in hindsight, we agree with the reviewer that this would be valuable. We now describe an additional protocol, Basic Protocol 3, which outlines how to generate and visualise a principal component analysis as a compositional biplot using the log-ratio transformed data produced by ALDEx2. This protocol uses the same input data as Basic Protocol 2 and the code is described fully (complete, annotated code available on GitHub in 'code/analysis_metatranscriptome.R').

7. In the critical parameters section under commetary - could you please add numbers of genes deemed significant / non-significant for the analysis shown in Figure 2B in the 2nd paragraph of the commentary text

We apologise for this omission and have added this information to the appropriate points in the corresponding paragraph.

8. Also for the 2nd paragraph - please explain how the scale difference figure of 14% was derived

We have now added a sentence to this paragraph explaining that steps 6 – 9 of Basic Protocol 2 outline how this value was reached. Basic Protocol 2 also has an italic note underneath step 9 describing this, as well.

9. In the section describing choice of gamma, it would be very helpful to have some specific recommendations or at least strategies that could be employed to define gamma e.g. performing GSEA on resultant genes to see which values provide optimal coherence in gene sets?? particularly when analysing 16S v RNASeq v metagenomics v metatranscriptomics - do more complex datasets require larger gammas for example?

We thank the reviewer for this comment as this is an important detail. At present, we can only offer specific advice on RNA-seq datasets: we initially covered this in the final paragraph of the Critical Parameters section and have expanded it according to point #11, adding a clearer explanation and explicit recommendations of what gamma values to choose and when. We would recommend against running multiple differential expression tools and comparing outputs as the false-discovery rates of several widely used tools are incredibly high and there is often limited overlap in genes which are reported as differentially expressed/abundant (see: https://doi.org/10.1101/2025.11.28.690516 and https://doi.org/10.1186/s13059-022-02648-4, respectively).

10. The 4th paragraph describes the scale parameter, I wonder if this belongs before the discussion of the gamma parameter.

We apologise for the confusion; we recognise our wording was not particularly clear. Gamma represents the degree of scale uncertainty being added to the overall model and we used these terms interchangeably at times without explaining them. We now clarify this explicitly at the beginning of the Critical Parameters section and have altered the paragraph in question to make this more evident.

11. The last part of the 4th paragraph describing the regression line needs a little more explanation, I did find this a little confusing.

We apologise for the confusing wording and hope that the changes made as part of the response to point 9 helps clear this up somewhat. We have separated this part of the paragraph into its own separate paragraph and have expanded slightly upon our intended meaning. We also offer the following, complete explanation to the reviewer:

For each gamma value from 0 to 1 (in increments of 0.1), we ran ALDEx2 on the yeast transcriptome dataset, for a total of 11 analyses. For each of these analyses, we find the smallest absolute value in the 'diff.btw' column of the ALDEx2 output (which represents the log difference between groups of a given feature) that also has a *P* value <0.05. This gives us an idea of the smallest log difference between two biological conditions that is required for ALDEx2 to be able to report a statistically significant result. Plotting these values against the corresponding values of gamma (i.e. the degree of scale uncertainty), we can see that increasing the gamma parameter of the aldex.clr() function requires there to be a higher log difference between conditions before it will recognise a feature as significantly different between groups. The slope of the regression line for these data points is approximately 2.3. This means that one can expect that using any given scale value to analyse this dataset will require a feature to have a difference between groups around 2.3 times as large as the scale value before ALDEx2 will recognise it as differentially expressed.

We are currently preparing another manuscript for submission where we repeat this kind of comparison (smallest log-difference required for a significant P value) 100 times across 10 different datasets with ALDEx2, including bulk transcriptome, metatranscriptome and single-cell transcriptome datasets. Remarkably, we found that the slopes for these different types of RNA-seq data are relatively constant; therefore, it appears that this guidance for ALDEx2 appears to be universal for RNA-seq data. We can sum up this finding as follows: to choose an appropriate gamma value, ask yourself how small of a difference between groups you personally consider to be biologically pertinent to your condition of interest. If you are only interested in relatively large differences, a gamma value between 0.4 and 0.5 will require a $\log_2$ fold-change of around 1.25 (i.e. ~2.4-fold difference between groups) before ALDEx2 will report a statistically significant result. Conversely, if you think that you will be dealing with more subtle differences, then a smaller gamma value is more appropriate- a gamma of 0.2 will require a log-fold change of around 0.5 (i.e. a ~1.45-fold difference between groups). Essentially, pick a gamma value and multiply it by 2.5: that's the minimum log2 fold-change between groups you will require for a feature to be considered significantly different between groups. Overall, if you expect a subtle difference between groups, use a smaller gamma value, whereas if you expect to see very large, obvious difference between groups, use a larger gamma value.

12. Table 2 should include the computer specs (I know it is in the main text, but it should also go in the table legend - also please add units for the time - seconds?)

We apologise for the accidental omission of the units. We have added units to the appropriate column and have added the machine specs to the legend of the table.

13. We did try running ALDEx2 with the GLM workflow to control for e.g. antibiotics use, but the scale modeling throws an error - this isn't included in the protocol and is an alternative approach, but perhaps the authors might want to include some additional commentary (might require a software fix?).

We apologise to the reviewer, but without seeing either the specific error, the actual code run, or the input data table and conditions vector, we cannot offer any comment or explanation for this error. The GLM function offered by ALDEx2, aldex.glm(), only requires an aldex.clr object, so as long as one is able to generate this, ALDEx2 should not have any issue, even when scale models are used. We can confirm that the aldex.glm() function runs without error in our hands when building a full, informed scale model from a publicly available metatranscriptome dataset, using the code below (should be fully reproducible on any machine with ALDEx2 v1.41 installed):

```
library(ALDEx2)

# pull KO-aggregated count table and metadata for vaginal metatranscriptomes
# used in Dos Santos et al. (2024)
url.df <-
"https://github.com/scottdossantos/dossantos2024study/raw/refs/heads/main/Rdata/ko.both.Rda"
tf.df <- tempfile(fileext = ".Rda")
download.file(url = url.df, destfile = tf.df, mode = "wb")
load(tf.df)
unlink(tf.df)

url.md <-
"https://github.com/scottdossantos/dossantos2024study/raw/refs/heads/main/Rdata/hm.metadata.Rda"
tf.md <- tempfile(fileext = ".Rda")
download.file(url = url.md, destfile = tf.md, mode = "wb")
load(tf.md)
unlink(tf.md)

# delete temp files
rm(list = c(ls(pattern = "url."),ls(pattern = "tf.")))

# remove two weird samples from the read count table and delete suspect reads
# of eukaryotic origin as in original paper
ko.both <- ko.both[,-c(9,22)]

euk <- paste("K03364","K13963","K01173","K12373", "K14327","K00863","K00599",
             "K13993","K00811","K03260","K00985", sep = "|")
```

```
ko.both <- ko.both[-which(grepl(euk, rownames(ko.both))),]

# set seed and generate matrix of (reproducible) scale values as in original
# paper
set.seed(2023)
mu.hbv <- aldex.makeScaleMatrix(gamma = 0.5, mu = c(1, 1.15), mc.samples = 128,
                                conditions = hm.metadata$`BV status`)

# make model matrix for a GLM including BV status (healthy vs. BV) and dataset
# (London or Europe datasets; this should have no impact on differential
# expression)
mm <- model.matrix(~ `BV status` + Dataset, hm.metadata)

# transform data w/ different amounts of scale added to each group (15 %
# difference between groups on average)
set.seed(2023)
x <- aldex.clr(reads = ko.both, conds = mm, mc.samples = 128,
               denom = "all", verbose = TRUE, gamma = mu.hbv)

# run the GLM function (will take 1-2 minutes to run)
x.glm <- aldex.glm(clr = x, verbose = T, fdr.method = "BH")

# calculate effect sizes from the GLM
x.glm.e <- aldex.glm.effect(clr = x, verbose = T, include.sample.summary = T)

# extract effect sizes and CLR abundances for both variables
effect.bv <- x.glm.e[[1]]
effect.dataset <- x.glm.e[[2]]

# analysis shows only a single significantly different gene expressed at a
# higher level in samples from the London dataset compared to Europe Dataset,
# which is expected because dataset shouldn't really influence this at all!
length(which(x.glm$`DatasetLondon:pval.padj` <0.05))

# look at this gene in particular: present in all London samples at low counts
# (~20-30 reads per sample) but absent from all Europe samples. This is very
# clearly an artifact of sequencing depth and not truly differential expression
ko.both[which(x.glm$`DatasetLondon:pval.padj` <0.05),]

# many differentially expressed genes based on BV status (note: original paper
# restricted analysis of differentially expressed genes to those with absolute
# effect size >1)
length(which(x.glm$`\`BV status\`Healthy:pval.padj` <0.05)) # 911
```

Given that ALDEx3 is on the horizon, we will be preparing a separate manuscript showing how one can run the comparisons as in the above example given by the reviewer. Using ALDEx3 would be preferred over the GLM module of ALDEx2 for two reasons: 1) ALDEx3 using a process known as streaming to achieve far superior speeds than ALDEx2; and 2) ALDEx3 is inherently based on linear models and the output will be easier to interpret than the current GLM module in its predecessor.

While the above code is provided for the reviewer only, we have included a short paragraph noting that ALDEx2 can implement GLMs that model scale uncertainty in the exact same manner as the more common aldex.effect() and aldex.ttest() functions.

14. In time considerations, can you mention if ALDEx2 supports e/g/ multithreading for cluster computing?

Briefly, yes but because of extensive optimisation of the ALDEx2 internals for vectorisation, and how R handles the overhead required for multithreading, it doesn’t offer much, if any, improvement in run time. For the sake of completeness, however, we have run the same 5 simulations on the three datasets in Table 2 using multi-core processing and have updated the Table accordingly.